

DOUALA, Cameroon

WMO: 649100

Lat: 4.006N

Lon: 9.719E

Elev: 10

StdP: 101.20

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.0	22.8	21.1	15.8	25.3	21.8	16.5	24.6	5.9	27.6	5.4	27.5	0.9	80	0.291

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.8	33.9	27.1	33.2	26.9	32.9	26.9	28.8	31.2	28.1	30.9	27.7	30.7	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.2	24.4	29.9	27.8	23.8	29.6	27.0	22.7	29.1	93.7	32.0	90.5	30.2	88.2	31.8	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.4	4.7	4.1	DB	21.1	35.2	0.7	0.5	20.6	35.5	20.2	35.8	19.9	36.1	19.4	36.5	(4)
(5)			WB	20.6	29.5	0.8	1.4	20.0	30.5	19.6	31.3	19.1	32.0	18.5	33.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	28.6	28.9	28.6	28.1	27.9	26.9	25.9	25.6	26.3	26.8	27.6	28.3	(6)	
	DBStd	1.46	0.79	1.01	1.12	1.23	1.13	1.00	0.96	0.87	1.00	1.00	1.12	0.77	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6369	576	528	575	543	555	507	494	485	489	522	527	568	(10)	
	CDD18.3	3327	317	295	317	293	297	257	236	227	239	263	277	309	(11)	
	CDH23.3	31002	3358	3307	3484	3043	3010	2242	1590	1334	1720	2139	2593	3183	(12)	
(13)	CDH26.7	10268	1312	1350	1415	1187	1098	574	238	146	338	584	848	1178	(13)	
(14)	Wind	WSAvg	1.3	1.2	1.4	1.4	1.3	1.3	1.3	1.4	1.3	1.3	1.2	1.0	(14)	
(15) Precipitation	PrecAvg	3815	34	56	170	242	274	422	644	724	608	402	138	35	(15)	
	PrecMax	4885	159	221	349	535	412	790	1208	1240	1027	632	354	124	(16)	
	PrecMin	2596	0	0	47	124	152	174	266	377	265	196	7	0	(17)	
	PrecStd	530	36	52	76	79	70	136	231	211	203	101	76	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	34.2	34.2	34.2	33.9	32.2	30.8	30.1	31.2	32.2	33.1	33.4	(19)	
		MCWB	26.3	27.0	27.4	27.5	27.3	26.6	26.0	25.8	26.1	26.6	27.1	26.5	(20)	
	2%	DB	33.2	33.8	33.6	33.2	33.0	31.2	29.8	29.0	30.2	31.2	32.2	33.0	(21)	
		MCWB	26.5	27.0	27.2	27.3	27.1	26.5	25.7	25.4	25.9	26.4	26.8	26.6	(22)	
	5%	DB	32.8	33.0	33.0	32.8	32.1	30.6	29.0	28.1	29.8	30.8	31.7	32.2	(23)	
		MCWB	26.5	26.9	27.1	27.2	26.8	26.2	25.3	25.1	25.7	26.3	26.7	26.5	(24)	
	10%	DB	32.0	32.2	32.1	31.9	31.2	29.8	28.1	27.2	28.9	29.9	30.9	31.8	(25)	
		MCWB	26.3	26.7	26.9	26.9	26.7	25.9	25.0	24.9	25.4	26.0	26.6	26.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.7	29.2	29.2	30.1	28.9	27.2	26.5	26.9	26.8	28.0	28.4	28.5	(27)	
		MCDB	31.0	31.5	31.4	31.7	31.6	30.6	29.2	28.0	29.6	29.6	30.5	31.0	(28)	
	2%	WB	27.8	28.2	28.2	28.5	27.8	26.7	26.0	26.0	26.3	27.0	27.6	27.4	(29)	
		MCDB	30.7	31.0	31.2	31.2	31.2	29.9	28.6	27.9	29.0	30.0	30.2	30.7	(30)	
	5%	WB	27.2	27.6	27.7	27.7	27.2	26.4	25.7	25.5	26.0	26.5	27.1	27.0	(31)	
		MCDB	30.5	30.6	30.9	30.8	30.6	29.4	28.0	27.3	28.6	29.4	30.3	30.6	(32)	
	10%	WB	26.8	27.2	27.2	27.2	26.8	26.1	25.4	25.1	25.6	26.1	26.7	26.7	(33)	
		MCDB	30.0	30.3	30.5	30.4	30.0	28.8	27.4	26.8	28.0	28.9	29.6	29.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	7.8	7.9	7.8	7.5	6.0	4.6	4.2	5.4	6.7	7.2	7.7	(35)	
		MCDBR	8.6	8.2	8.6	8.8	8.4	7.2	5.8	5.2	6.7	7.9	8.0	8.1	(36)	
	5% WB	MCWBR	2.9	2.8	3.4	3.7	3.6	3.0	2.4	2.3	2.8	3.5	3.3	2.9	(37)	
		MCWBR	8.0	7.9	8.1	8.2	7.9	6.5	5.1	4.7	5.9	7.2	7.5	7.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.803	0.832	0.805	0.570	0.503	0.524	0.522	0.507	0.450	0.467	0.587	0.739	(40)	
		taud	1.525	1.495	1.543	2.016	2.213	2.141	2.146	2.193	2.369	2.335	1.950	1.623	(41)	
	Ebn at Noon		597	595	615	763	796	767	773	802	865	855	747	629	(42)	
		Edh at Noon	294	308	294	180	143	151	152	148	127	131	190	262	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.05	4.79	4.63	4.68	4.53	3.77	3.30	3.19	3.69	4.11	4.42	4.85	(44)	
	RadStd		0.26	0.29	0.28	0.15	0.20	0.34	0.28	0.21	0.22	0.17	0.18	0.17	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page